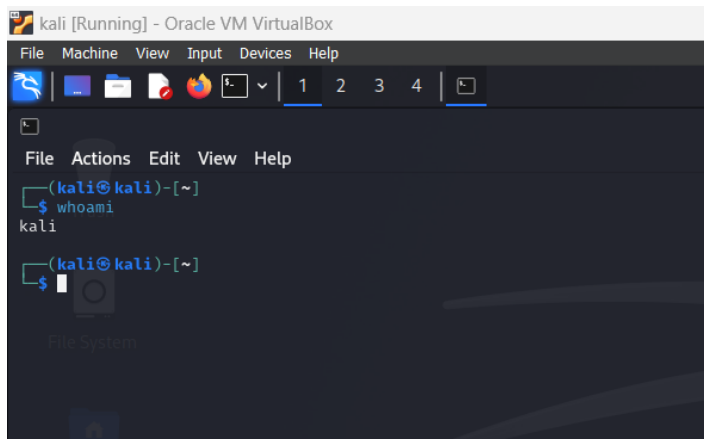


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Section A – Linux Basics

Current username



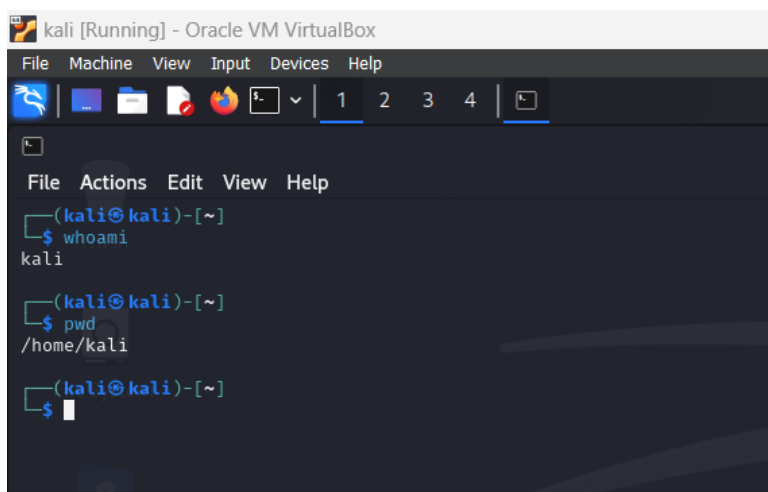
```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
(kali@kali)-[~]
$ whoami
kali
(kali@kali)-[~]
```

Command used: whoami

It is useful for:

- **Verifies Current Identity:** Displays the exact active username to check if you are logged in as a normal user or root/administrator.
- **Automates Script Security:** Allows bash and batch scripts to check user identity and block non-authorized accounts from running sensitive commands.
- **Audits System Privileges:** Helps troubleshoot "Permission Denied" errors and, on Windows (whoami /priv), lists all active security group permissions.

Current working directory



```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
(kali@kali)-[~]
$ whoami
kali
(kali@kali)-[~]
$ pwd
/home/kali
(kali@kali)-[~]
$
```

Command used: pwd

- A working directory is the specific folder where your terminal or program is currently looking
- Any file you create, open, or delete without typing a full path happens inside this folder.

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Creating a folder named CyberLab.

```
(kali㉿kali)-[~]
└─$ mkdir CyberLab

(kali㉿kali)-[~]
└─$ tree
.
├── CyberLab
├── Desktop
│   ├── hash.txt
│   ├── tash.txt
│   └── yash.txt
├── Documents
├── Downloads
├── Music
├── Pictures
├── Public
└── .
```

Command used: mkdir

Difference between a file and a directory is:

- A file holds actual data, not any folder or directory while a directory is a container used to organize files and other directories.
- A file holds content and data inside it. whereas a directory contains files and subdirectories.

Navigating inside the CyberLab folder and moving back!

```
├── Videos
├── physics
├── yash
│   ├── c
│   ├── chemistry
│   └── physics
└── .

15 directories, 3 files

(kali㉿kali)-[~]
└─$ cd CyberLab

(kali㉿kali)-[~/CyberLab]
└─$ cd ..

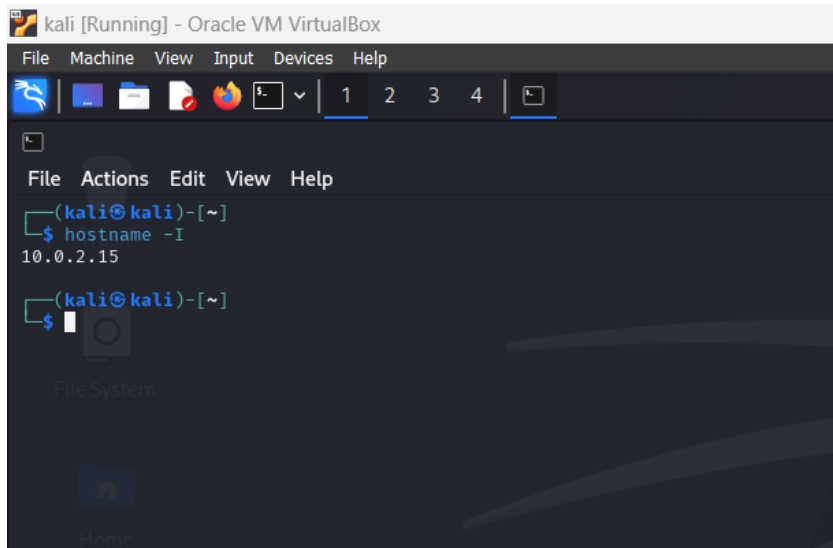
(kali㉿kali)-[~]
└─$
```

- **Command used to move inside this folder is: cd CyberLab**
it basically changes directory from the available options
- **Command used to move back is: cd ..**
it brings you out of the current directory

Section B – Network Identification

Finding my IP address. Finding if it is public or private.

Name: Yash Shrivastava

A screenshot of a Kali Linux terminal window titled 'kali [Running] - Oracle VM VirtualBox'. The terminal shows the command 'hostname -I' being executed, which returns the IP address '10.0.2.15'. The terminal interface includes a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help', and a toolbar with various icons. The prompt is '(kali@kali)-[~]'.

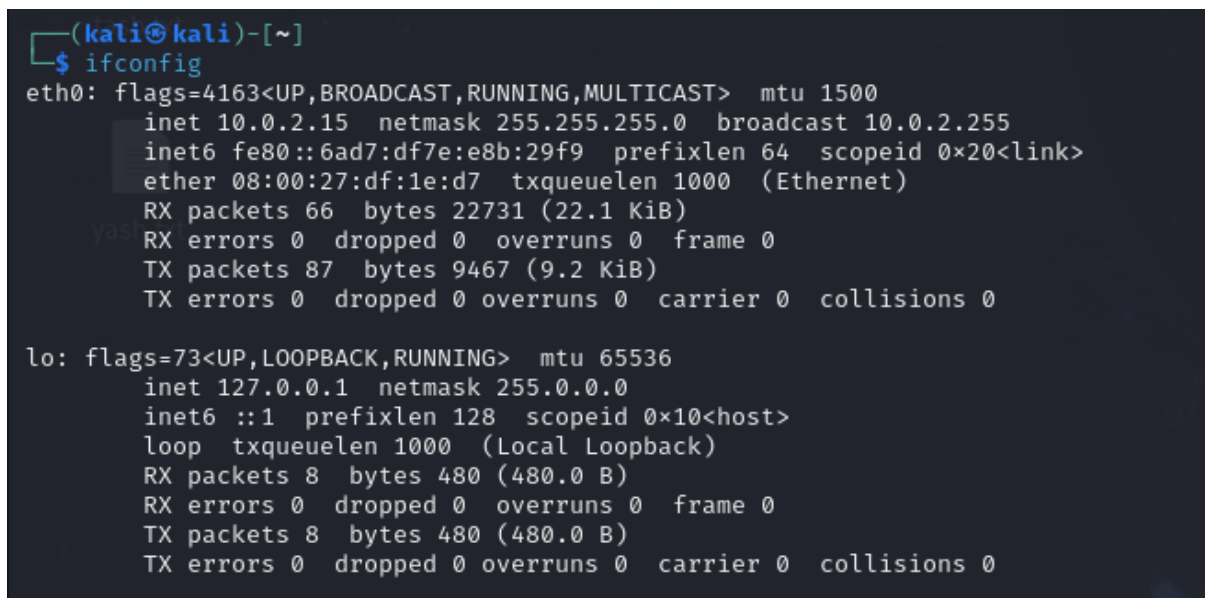
Command: `hostname -I`

It is a private IPv4 address, because it falls in these given range:

- **10.X.X.X** (e.g., 10.0.0.1)
- **172.16.X.X to 172.31.X.X** (e.g., 172.16.0.5)
- **192.168.X.X** (e.g., 192.168.1.15)

When an IP address falls out of these range, then it is a public IP address.

Identifying my subnet

A screenshot of a Kali Linux terminal window showing the output of the 'ifconfig' command. The output displays details for the 'eth0' interface, including its IP address (10.0.2.15), netmask (255.255.255.0), and broadcast address (10.0.2.255). It also shows statistics for RX and TX packets and bytes. The 'lo' interface (loopback) is also shown with its IP address (127.0.0.1) and netmask (255.0.0.0). The terminal prompt is '(kali@kali)-[~]'.

Command: `ifconfig`

Netmask: 255.255.255.0

Subnetting is the process of dividing a single large network into smaller, manageable sub-networks (subnets) to improve speed, efficiency, and security.

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Why Subnetting is Essential

- **Reduces Traffic:** Broadcast traffic stays inside its own subnet instead of slowing down the entire network.
- **Boosts Security:** It isolates sensitive departments (like Finance) from general network access.

network interfaces and identifying the active interface.

```
(kali㉿kali)-[~]
$ ip r
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:df:1e:d7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84881sec preferred_lft 84881sec
    inet6 fe80::6ad7:df7e:e8b:29f9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- Command: ip r
Displays the routing table, showing the paths and rules data packets follow to reach external networks.
- Command: ip a
Displays all network interfaces, their physical status, and their assigned IP addresses.
- Difference between ip r and ip a
ip a tells you who you are on the network, while ip r tells you where to send your data.

Section C – Information Gathering

Finding the IP address of google.com.

```
(kali㉿kali)-[~]
$ nslookup google.com
Server:      10.168.216.157
Address:     10.168.216.157#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.193.14
Name:   google.com
Address: 2404:6800:4002:827::200e
```

Command: nslookup google.com

Name: Yash Shrivastava

nslookup allows us to manually query DNS servers to test DNS resolution, find IP addresses, or troubleshoot network issues

DNS (Domain Name System) resolution is the process of converting a human-readable website name (like google.com) into a machine-readable IP address (like 142.250.190.46). It acts as the phonebook of the internet because routers require numerical IP addresses to fetch web pages.

Finding the mail server records for google.com

```
(kali@kali)-[~]
$ dig google.com

; <<>> DiG 9.20.0-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 56430
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 260     IN      A      142.250.193.14

;; Query time: 295 msec
;; SERVER: 10.168.216.157#53(10.168.216.157) (UDP)
;; WHEN: Mon Jun 22 14:40:33 EDT 2026
;; MSG SIZE rcvd: 55
```

The dig (Domain Information Groper) command is the modern, professional replacement for nslookup because it provides cleaner formatting, more detailed technical data, and follows standard DNS reply formats.

How to read it:

- **google.com.:** The domain name you queried.
- **260:** The TTL (Time to Live) in seconds, showing how long this result can be cached.
- **IN:** Internet class.
- **A:** The record type (A record maps a hostname to an IPv4 address).
- **142.250.193.14:** The final resolved IP address.

MX (Mail Exchanger) records are specific settings in a domain's DNS configuration that specify which mail servers are responsible for accepting incoming email messages on behalf of that domain.

Without an MX record, external email servers (like Gmail or Yahoo) will not know where to route emails sent to an address

example: user@yourdomain.com.

Performing a WHOIS lookup on github.com.

Name: Yash Shrivastava

```
(kali㉿kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-06-22T19:51:15Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
```

Name: Yash Shrivastava

```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: github.com
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:33+0000
Creation Date: 2007-10-09T18:20:50+0000
Registrar Registration Expiration Date: 2026-10-09T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Registrant Organization: GitHub, Inc.
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Name Server: dns2.p08.nsone.net
Name Server: dns4.p08.nsone.net
Name Server: ns-520.awsdns-01.net
Name Server: dns1.p08.nsone.net
Name Server: ns-1283.awsdns-32.org
Name Server: ns-1707.awsdns-21.co.uk
Name Server: dns3.p08.nsone.net
Name Server: ns-421.awsdns-52.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-06-22T19:50:14+0000 <<<

For more information on WHOIS status codes, please visit:
https://www.icann.org/resources/pages/epp-status-codes

If you wish to contact this domain's Registrant or Technical
contact, and such email address is not visible above, you may do so via our web
form, pursuant to ICANN's Temporary Specification. To verify that you are not a
robot, please enter your email address to receive a link to a page that
facilitates email communication with the relevant contact(s).

Web-based WHOIS:
https://domains.markmonitor.com/whois/contact/github.com

If you have a legitimate interest in viewing the non-public WHOIS details, send
your request and the reasons for your request to whoisrequest@markmonitor.com
and specify the domain name in the subject line. We will review that request and
may ask for supporting documentation and explanation.
```

Command: whois github.com

The **whois command** is a command-line tool used to look up ownership, registration, and contact information for a registered domain name or an IP address block.

It queries global WHOIS databases to tell you exactly who owns a piece of internet real estate

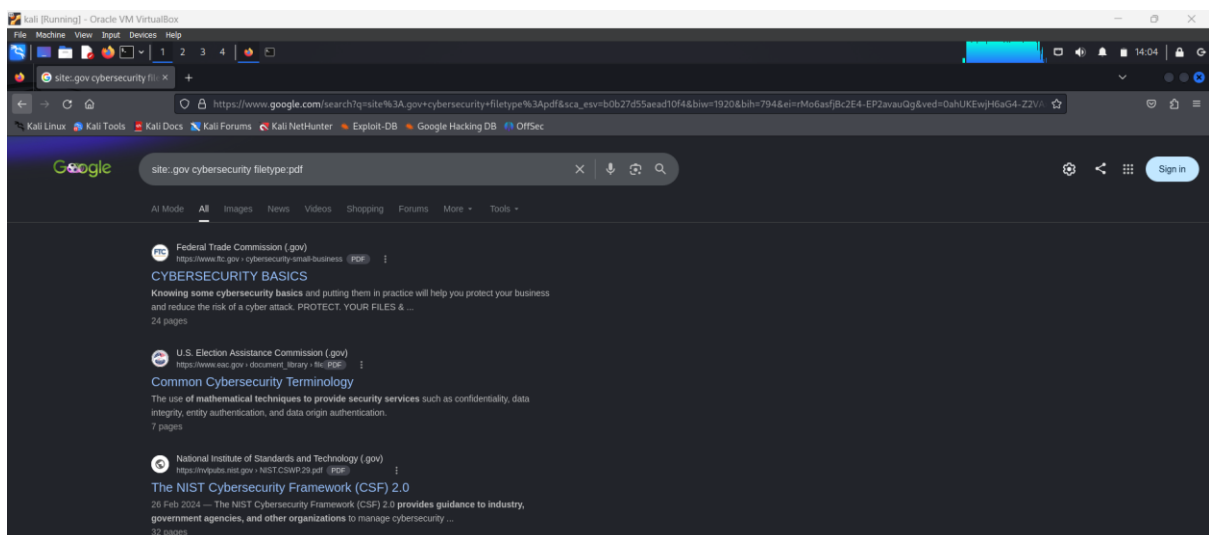
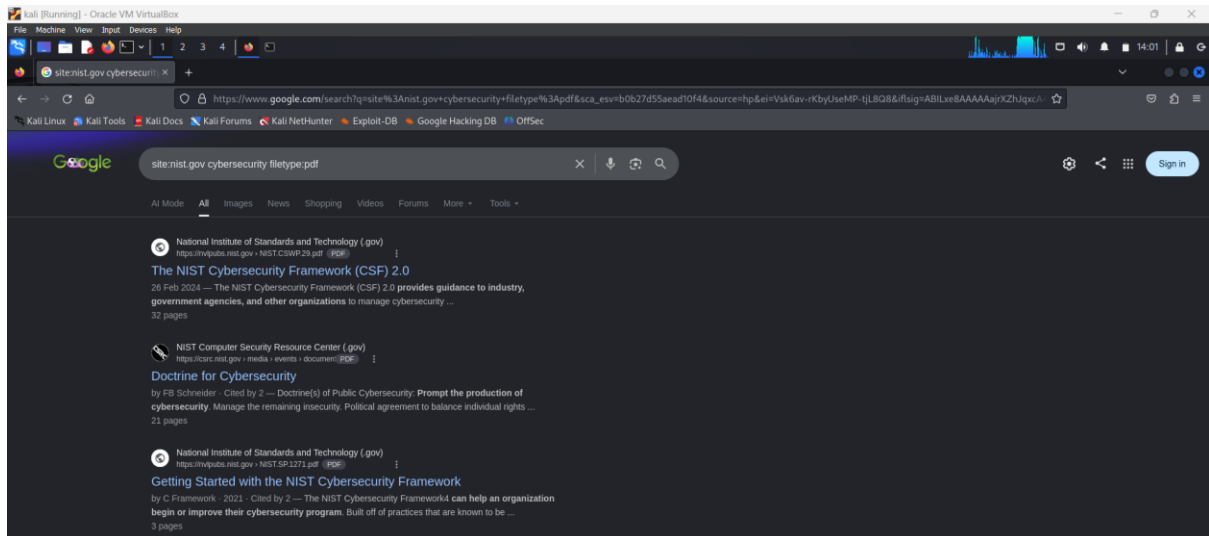
Information Provided by WHOIS

- Domain name
- Registration date
- Expiry date
- Domain status
- Name servers (DNS servers)
- Registrant/organization details (if publicly available)
- Administrative and technical contact information

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Section D – Google Dorking

Finding PDF documents related to cybersecurity.



Command: `site:.gov cybersecurity filetype:pdf`,

- `site:.gov`: This restricts your search exclusively to websites that use the .gov top-level domain. This ensures that every single link returned belongs to an official government agency (like the FTC, EAC, or NIST), completely filtering out blogs, commercial sites, or forums.
- `cybersecurity`: This acts as your core keyword. Google looks for web pages or documents where the term "cybersecurity" explicitly appears in the title, URL, or body text.
- `filetype:pdf`: This forces Google to only return direct links to downloadable PDF documents. It filters out standard HTML web pages, images, and other file types, allowing you to instantly locate research papers, frameworks, and official guides.

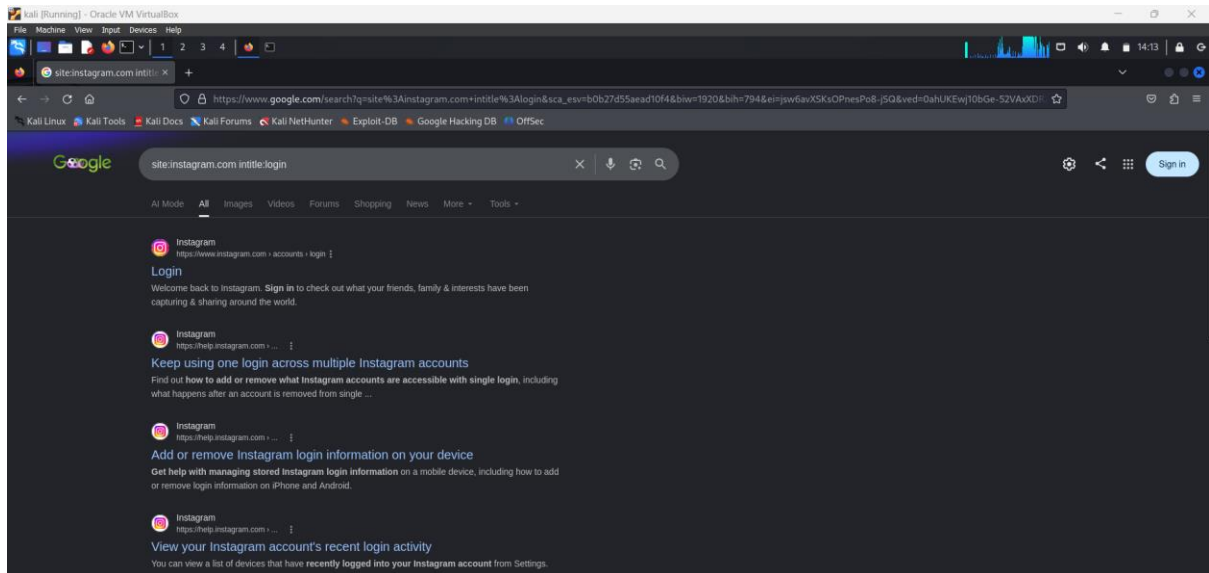
Relevant PDF documents found during scan

Documents uncovered:

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- **CYBERSECURITY BASICS** by the Federal Trade Commission (FTC)
- **Common Cybersecurity Terminology** by the U.S. Election Assistance Commission (EAC)
- **The NIST Cybersecurity Framework (CSF) 2.0** by the National Institute of Standards and Technology (NIST)

Searching for login pages on a domain.

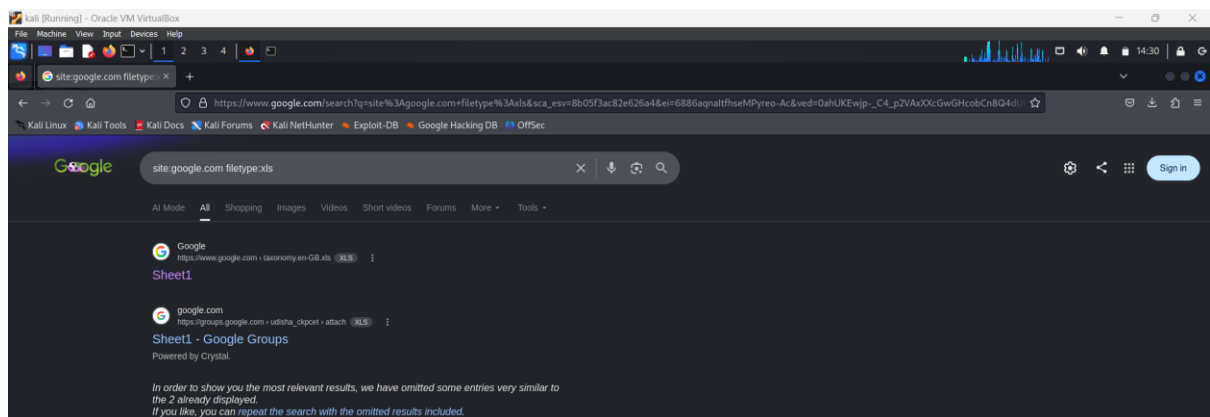
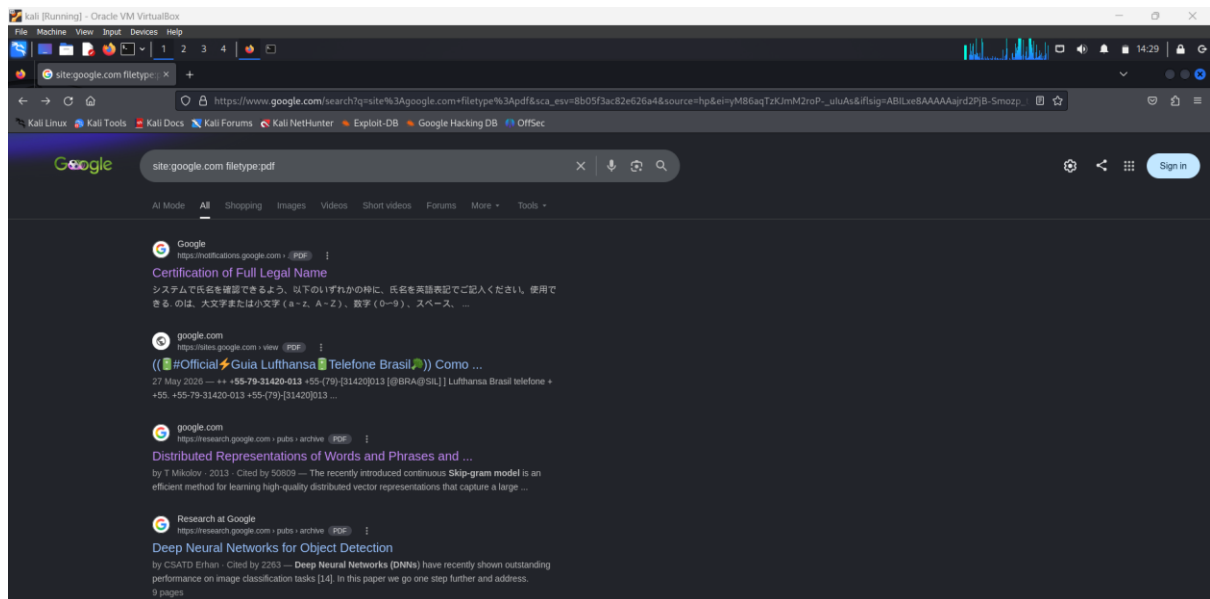


Your current methodology relies on filtering pages based on their structural HTML properties and domain ownership:

1. **Domain Isolation (site:instagram.com):** This narrows down the entire internet to look *only* at URLs containing the instagram.com domain. It completely filters out third-party phishers, blogs discussing Instagram, or news articles.
2. **HTML Title Probing (intitle:login):** This instructs Google to scan the meta-title tags <title> embedded in the HTML header of the target webpages. Web developers almost universally name their user authentication interfaces with clear descriptors like "Login", "Sign In", or "Log On". By pairing this with the site operator, you instantly pinpointed the platform's primary authentication page (://instagram.com) alongside its associated official login help documentation.

Finding publicly available documents on a target website.

Name: Yash Shrivastava



Command: site:google.com filetype:pdf

Command: site:google.com filetype:xls

Section E – Nmap Practical

Perform a basic scan against scanme.nmap.org.

Name: Yash Shrivastava

```
(kali㉿kali)-[~]
$ nmap scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-22 16:00 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.45s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
110/tcp   open  pop3
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 46.54 seconds
```

A basic Nmap scan of scanme.nmap.org (74.207.244.221) confirms the host is online and reveals open SSH (22), HTTP (80), nping-echo (9929), and Elite (31337) ports, alongside filtered ports indicating firewall activity.

The scan, which evaluates the top 1,000 TCP ports, highlights a small, intentionally accessible attack surface for practice

Identifying all open ports.

```
(kali㉿kali)-[~]
$ nmap -Pn 445 10.10.232.162
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-23 06:06 EDT
Nmap scan report for 445 (0.0.1.189)
Host is up (0.069s latency).
Not shown: 973 filtered tcp ports (net-unreach), 26 filtered tcp ports (no-response)
PORT      STATE SERVICE
110/tcp   open  pop3

Nmap scan report for 10.10.232.162
Host is up.
All 1000 scanned ports on 10.10.232.162 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 2 IP addresses (2 hosts up) scanned in 30.46 seconds
```

Command used: nmap -Pn 445 10.10.232.162

Port: 110

Protocol: TCP (unencrypted Transmission Control Protocol)

Used for: Post Office Protocol version 3 (POP3)

State: Filtered

How it Works

- **Message Retrieval:** POP3 downloads your emails directly to your local device and typically deletes them from the server.
- **Unencrypted:** Port 110 transmits data, including your username and password, in plain text. Because of this vulnerability to eavesdropping and man-in-the-middle attacks, it is highly recommended to use the secure, encrypted port **995 (POP3S)** instead

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Performing service version detection.

```
(kali㉿kali)-[~]  
$ nmap -sV 10.0.2.15  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-23 06:14 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.00042s latency).  
All 1000 scanned ports on 10.0.2.15 are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.38 seconds
```

Command used: nmap -sV 10.0.2.15

Why Service Version Information Matters

Discovering that a port is simply "open" (e.g., port 80/HTTP) only reveals a general entry point. Version detection is critical for network security due to three core mechanisms:

1. Pinpointing Exploitable Vulnerabilities (Mapping CVEs)

Vulnerabilities are almost always specific to exact software versions. For example, knowing a host runs an HTTP service is unhelpful on its own. However, if version detection reveals it runs Apache httpd 2.4.49, a security analyst immediately knows it is vulnerable to a severe path traversal and remote code execution flaw ([CVE-2021-41773](#)). It allows you to match the system's exact inventory footprint against global vulnerability databases like the National Vulnerability Database (NVD).

2. Defeating "Security through Obscurity"

Many network administrators change standard application ports to trick basic network scans (e.g., running an unencrypted Telnet or SSH management console on Port 80 instead of Port 22 or 23). Nmap's -sV scan does not guess based on the port number. It evaluates application banners, protocol handshakes, and behavioral response logic to identify the exact service name regardless of what port it uses.

3. Streamlining Automation and Patch Verification

Detailed version information enables blue teams to automate patch management checks. Instead of manually auditing configuration files on every virtual container or server, a simple internal version scan quickly maps out-of-date assets that require rapid remediation.

Performing host discovery on your local network and identifying likely devices.

Name: Yash Shrivastava

```
(kali㉿kali)-[~]  
$ nmap -sn 10.0.2.15/24  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-23 13:50 EDT  
Nmap scan report for 10.0.2.59  
Host is up (0.21s latency).  
Nmap scan report for 10.0.2.115  
Host is up (0.21s latency).  
Nmap scan report for 10.0.2.136  
Host is up (0.41s latency).  
Nmap scan report for 10.0.2.157  
Host is up (0.81s latency).  
Nmap scan report for 10.0.2.209  
Host is up (0.44s latency).  
Nmap done: 256 IP addresses (5 hosts up) scanned in 51.71 seconds
```

Command: 10.0.2.15/24

10.0.2.15 (Implicit Network Gateway): While not explicitly listed as up in this truncated terminal view, in a standard 10.0.2.x virtualization subnet (like VirtualBox NAT Network), .2 is almost always reserved for the virtual default gateway/router interface.

Your Active Lab Targets (10.0.2.59, 10.0.2.115, 10.0.2.136, 10.0.2.157, 10.0.2.209): These 5 IP addresses represent active machines running on this internal network. In a typical cybersecurity lab or classroom environment, these are highly likely to be:

- **Metasploitable VMs:** Vulnerable Linux or Windows machines meant for penetration testing practice.
- **Windows Server / Active Directory Targets:** Common target nodes in multi-machine local network labs.
- **Other Student/User Environments:** If you are sharing a NAT Network or Host-Only network layout with other virtual machines.